

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
3	(a)			A: nucleus (1) Accept nucleolus B: chloroplast (1)	1	1		2		
	(b)			apparent length = $13\text{mm} / 13 \times 1000 = 13000\mu\text{m}$ magnification = $13000 / 32.3 \mu\text{m}$ or $13/0.0323$ (1) magnification = $402.48/ 402.5/ 403/ 402$ (correct answer = 2 marks) OR apparent length = $13.5\text{mm} / 13.5 \times 1000 = 13500\mu\text{m}$ magnification = $13500 / 32.3 \mu\text{m}$ or $13.5/ 0.0323$ (1) magnification = $417.96/ 418$ (correct answer = 2 marks) Award 1 mark for evidence of figures showing image size/ actual size		2		2	2	
	(c)			1. straight chains of β-glucose/ alternative molecules of β-glucose rotate through 180° / chains cross-linked / form microfibrils (1) 2. which provide {strength/rigidity/inelasticity} to the cell wall (1) support = neutral 3. When the {solute concentration/ solute potential} changes (water potential will change) causing water to {move into /out of} the cell (1) 4. cell wall prevents {osmotic lysis/ cell bursting}/ cell wall prevents cell shrinking (1) turgid/ plasmolysed = neutral Reject if direction of water movement is incorrect	1 1		1 1	4		

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3	(d)			<i>Spirogyra</i> – eukaryotic cells & <i>Nostoc</i> - prokaryotic cells (1) Similarity: Both contain ribosomes / cell membranes / DNA/ genetic material (1) Difference: any 2 from: (1 mark) <table><tr><th><i>Spirogyra</i></th><th><i>Nostoc</i></th></tr><tr><td>{Membrane bound organelles/ named organelle} present</td><td>{Membrane bound organelles/ named organelle} absent</td></tr><tr><td>DNA enclosed within nuclear membrane</td><td>DNA free in cytoplasm</td></tr><tr><td>Linear DNA</td><td>Loop of DNA Accept plasmid</td></tr><tr><td>Larger/80s ribosomes Accept ribosomes are different sizes</td><td>Smaller/70s ribosomes</td></tr><tr><td colspan="2">Accept following</td></tr><tr><td>DNA associated with histones</td><td>DNA not associated with histones</td></tr><tr><td>Mesosome absent</td><td>Mesosome present</td></tr></table>	<i>Spirogyra</i>	<i>Nostoc</i>	{Membrane bound organelles/ named organelle} present	{Membrane bound organelles/ named organelle} absent	DNA enclosed within nuclear membrane	DNA free in cytoplasm	Linear DNA	Loop of DNA Accept plasmid	Larger/80s ribosomes Accept ribosomes are different sizes	Smaller/70s ribosomes	Accept following		DNA associated with histones	DNA not associated with histones	Mesosome absent	Mesosome present		1 1 1		3		
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				Question 3 total	3	6	2	11	2	0																